

CAT Algebra – Mixed Practice Set 1 of 3

Topics: Polynomials · Quadratics · Binomial · Inequalities · Functions · GIF · Modulus · Logarithms · Progressions · Co-ordinate Geometry · Maxima & Minima

QUESTIONS

Q1.

[Polynomials]

If $x=1, 2, -3$ are three roots of $P(x)=Ax^4+Bx^3+Cx^2+Dx+E$, how many of the following are always true?

- A. $P(2)=0$
- B. $P(x)=(x-1)(x-2)(x+3)$
- C. 4th root could be real or imaginary.
- D. If $E=12$ and $A=2$, the 4th root is -1 .

- (1) One
- (2) Two
- (3) Three
- (4) Four

Q2.

[Polynomials]

If $f(x)=x^3+3x^2+3x$, find $f(x+1)$.

- (1) x^3+4x^2+5
- (2) $2x^3+6x^2+3$
- (3) $3x^3+12x^2+4x+1$
- (4) $x^3+6x^2+12x+7$

Q3.

[Polynomials]

If $(x+a)$ is a factor of x^3+ax^2+ax+4 , the values of a are:

- (1) 2
- (2) -2
- (3) Either
- (4) Neither

Q4.

[Polynomials]

If a, b, c are real with $a^2+b^2+2c^2-4a+2c-2bc+5=0$, find $a+b-c$.

- (1) 0
- (2) 2
- (3) -2
- (4) 4

Q5.

[Linear Equations]

A thief gave each of 3 watchmen half his diamonds plus 2, and escaped with 1. How many did he steal?

- (1) 40
- (2) 36
- (3) 25
- (4) None of these

Q6.

[Quadratic Equations]

Ujagar got roots (4,3) with wrong constant; Keshab got roots (3,2) with wrong x-coefficient. Actual roots:

- (1) (6,1)
- (2) (-3,-4)
- (3) (4,3)
- (4) (-4,-3)

Q7.

[Quadratic Equations]

If sum of roots of $x^2+ax+1=0$ equals sum of squares of reciprocals of roots, a possible value of a is:

- (1) -1
- (2) 2
- (3) 1
- (4) 4

Q8.

[Quadratic Equations]

Find the value of $\sqrt{6-\sqrt{6-\sqrt{6-\dots\infty}}}$.

- (1) 3
- (2) 2
- (3) 2.5
- (4) $\sqrt{6}$

Q9.

[Quadratic Equations]

A quadratic $f(x)$ has maximum 3 at $x=1$ and $f(0)=1$. Find $f(10)$. (CAT 2007)

- (1) -159
- (2) -110
- (3) -180
- (4) -105

Q10.

[Higher Degree]

m is the smallest positive integer such that $n^3-7n^2+11n-5 > 0$ for all $n \geq m$. (CAT 2001)

- (1) 4
- (2) 5
- (3) 8
- (4) None of these

Q11.

[Binomial]

How many terms in $(a-b)^{30}$ are negative (a, b positive)?

- (1) 0
- (2) 1
- (3) 15
- (4) 30

Q12.

[Binomial]

The coefficient of x^7 in $(1-x^2+x^3)(1+x)^{10}$ is:

- (1) 75
- (2) 78
- (3) 85
- (4) None of these

Q13.

[Inequality]

Which values of P satisfy $P(P-3) < 4P-12$?

- (1) $P > 13$
- (2) $24 \leq P < 71$
- (3) $3 < P < 4$
- (4) None of these

Q14.

[Inequality]

How many integers a satisfy $9x^2 + 3ax + (a+5) > 0$ for all real x ?

- (1) 7
- (2) 8
- (3) 9
- (4) 10

Q15.

[Inequality]

If $x < y$, which is always true?

- (1) $x < (x+y)/2 < y$
- (2) $x < \sqrt{xy} < y$
- (3) $x < y^2 - x^2 < y$
- (4) $x < xy < y$

ANSWER KEY

Q1: (2) Two **Q2:** (4) $x^3+6x^2+12x+7$ **Q3:** (3) Either **Q4:** (2) 2 **Q5:** (2) 36
Q6: (1) (6,1) **Q7:** (3) 1 **Q8:** (2) 2 **Q9:** (1) -159 **Q10:** (4) None of these
Q11: (3) 15 **Q12:** (2) 78 **Q13:** (3) $3 < P < 4$ **Q14:** (3) 9 **Q15:** (1) $x < (x+y)/2 < y$

DETAILED SOLUTIONS

Solution Q1 — [Polynomials]

Answer: (2) Two

A is true (2 is a root). B is false (needs $A=1$ and the 4th root). C is false (imaginary roots come in pairs). D is true: product of roots $= E/A = 6$, so $1 \times 2 \times (-3) \times r = 6 \rightarrow r = -1$. Statements A and D $\rightarrow 2$.

Solution Q2 — [Polynomials]

Answer: (4) $x^3+6x^2+12x+7$

Expand $(x+1)^3 + 3(x+1)^2 + 3(x+1)$ to get $x^3+6x^2+12x+7$.

Solution Q3 — [Polynomials]

Answer: (3) Either

Substituting $x=-a$ gives $-a^3+a^3-a^2+4=0 \rightarrow a^2=4 \rightarrow a=\pm 2$.

Solution Q4 — [Polynomials]

Answer: (2) 2

LHS $= (a-2)^2 + (c-b)^2 + (c+1)^2 = 0 \rightarrow a=2, b=-1, c=-1$. So $a+b-c=2$.

Solution Q5 — [Linear Equations]

Answer: (2) 36

Backwards: before 3rd watchman: $(1+2) \times 2 = 6$. Before 2nd: $(6+2) \times 2 = 16$. Before 1st: $(16+2) \times 2 = 36$.

Solution Q6 — [Quadratic Equations]

Answer: (1) (6,1)

Sum of roots $= 7$ (from Ujagar, correct). Product $= 6$ (from Keshab, correct). Equation: $x^2-7x+6=0 \rightarrow$ roots (6,1).

Solution Q7 — [Quadratic Equations]

Answer: (3) 1

Setting $-a = a^2 - 2$ gives $a^2 + a - 2 = 0 \rightarrow a = 1$ or -2 .

Solution Q8 — [Quadratic Equations]

Answer: (2) 2

Let $N = \sqrt{6-N}$. Then $N^2 + N - 6 = 0 \rightarrow N = 2$ (positive root).

Solution Q9 — [Quadratic Equations]

Answer: (1) -159

$f(x) = a(x-1)^2 + 3$. $f(0) = a + 3 = 1 \rightarrow a = -2$. $f(10) = -2(81) + 3 = -159$.

Solution Q10 — [Higher Degree]

Answer: (4) None of these

$n^3 - 7n^2 + 11n - 5 = (n-1)^2(n-5)$. Zero at $n=5$, positive for $n \geq 6$. So $m=6$.

Solution Q11 — [Binomial]

Answer: (3) 15

Odd powers of b give negative terms: powers 1,3,5,...,29 $\rightarrow$ 15 terms.

Solution Q12 — [Binomial]

Answer: (2) 78

$${}^{10}C_7 - {}^{10}C_5 + {}^{10}C_4 = 120 - 252 + 210 = 78.$$

Solution Q13 — [Inequality]

Answer: (3) $3 < P < 4$

$$P(P-3) < 4(P-3) \rightarrow (P-3)(P-4) < 0 \rightarrow 3 < P < 4.$$

Solution Q14 — [Inequality]

Answer: (3) 9

$$\text{Discriminant} < 0: a^2 - 4a - 20 < 0 \rightarrow -2.89 < a < 6.89. \text{ Integer values: } -2 \text{ to } 6 \rightarrow 9.$$

Solution Q15 — [Inequality]

Answer: (1) $x < (x+y)/2 < y$

The AM always lies strictly between x and y when $x < y$.
